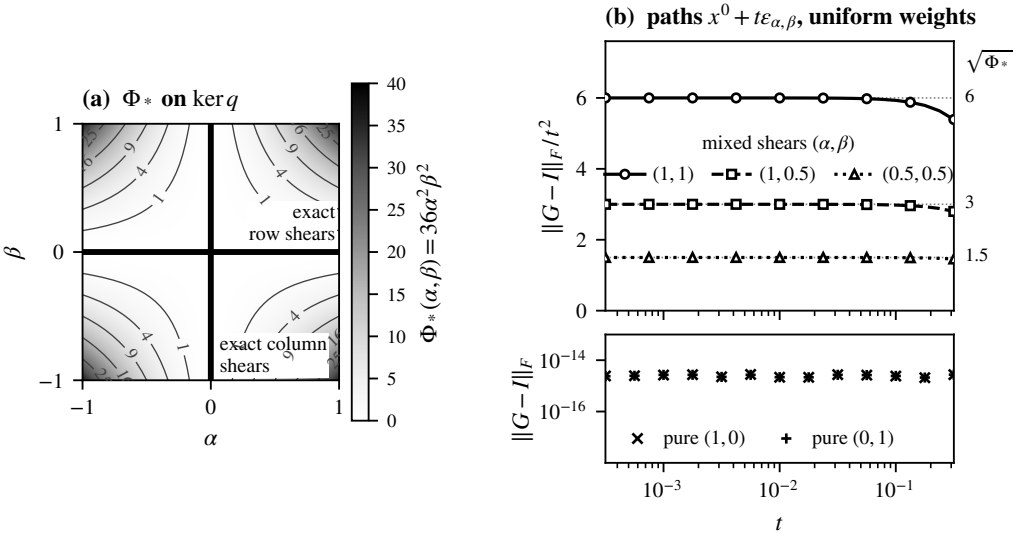
(a) Φ_* on $\ker q$
exact row shears
exact column shears
α
β
$\Phi_*(\alpha,\beta) = 36\alpha^2\beta^2$
(b) paths $x^0 + t\varepsilon_{\alpha,\beta}$, uniform weights
mixed shears (α,β)
(1, 1)
(1, 0.5)
(0.5, 0.5)
$\sqrt{\Phi_*}$
$\|G - I\|_F / t^2$
pure (1, 0)
pure (0, 1)
$\|G - I\|_F$
t